

**PROGRAMME: BCA (Bachelor of Computer Applications)- MACT
SEMESTER – IV**

Teaching-Learning & Evaluation Plan

Course Information: CORE COURSES

Course Code: 22BCA4M01 Course Title: **Mobile Computing**

Credits Units: **03**

Contact Hours: **45**

L-T-P: **3-0-0**

IA: ESE Weightage - **50: 50**

Pass Marks (IA & ESE) –**NA : 18**

Aggregate Pass Marks: **40**

ESE Question Paper Marks: **50**

Special Examination Fees: **NA**

Pre-requisite (if any):

1. Students have completed course on computer programming.

Course Facilitator (s): **Dr. Taskeen Zaidi**

Programme Outcomes (POs)

At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

Program Specific Outcomes (PSO's) MACT

PSO 1	Discuss the basic knowledge of Cloud Computing and Android Technology.
PSO 2	Articulate the knowledge on various cloud and mobile service providers.
PSO 3	Demonstrate latest tools, techniques and resources for various cloud and mobile computing activities
PSO 4	Explore solutions for real-time problems in Mobile Applications and Cloud Technology.

Course Objectives (COB's)

COB1: To study the specifications and functionalities of various protocols/standards of mobile networks.

COB2: To discuss various types of layouts and services and user interface design

COB3: To gain knowledge about different mobile platforms and application development

Course Outcomes (CO's)

At the end of the course, students will be able to

Sl. No.	Course Outcome	Description	Bloom's Taxonomy Level
1.	CO 1	Describe mobile communication technologies and telecommunication system.	L2
2.	CO 2	Compare Mobile computing and wireless networking.	L2
3.	CO 3	Articulate Android application development environment.	L3
4.	CO 4	Illustrate Android activities and GUI design Concepts.	L3
5.	CO 5	Explore User Interface design on Android platform.	L4

CO-PO/PSO Mapping: MACT

	Blooms Taxonomy Levels	CO-PO Mapping (3-strong, 2-Good, 1-Weak)												Program Specific Outcomes (PSO's)			
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
														L2	L3	L3	L4
CO1	L2	3	1	1	-	-	-	2	1	1	-	-	-	2	1	-	-
CO2	L2	2	1	1	-	-	-	2	1	1	-	-	-	2	2	-	-
CO3	L3	2	3	3	1	2	-	3	1	1	-	1	-	2	2	3	2
CO4	L3	2	3	3	1	2	-	2	2	1	-	1	-	2	3	3	2
CO5	L4	3	3	3	1	3	-	1	2	1	-	1	-	2	2	3	3

Course Contents:

Module No	Contents	Tools used/Assessment and Activity	CO Mapping	PO Mapping
Module 1 (9Hours)	Module 1: Basics of Communication Technologies Mobile handsets, Wireless Communications and Server Applications, Cell phone System, Types of Telecommunication Networks, Components of wireless communication system, Architecture of mobile telecommunication system, wireless networking standards, Wireless LANs, Wireless LAN Architecture, Applications of WLANs, Advantages of WLANs over wired LANs, Bluetooth Technology, Protocol stack of Bluetooth.	Activity1: Online Certification Course – 13 to 15 Hours Linked in Learning Internal Test-1 Preparatory Test	CO1	PO1,PO7,PO9
Module 2 (9Hours)	Module 2:Mobility Support in IP and TCP Mobile IP – Mobile Agent, Foreign Agent, Care of Address, Registration, Advertisement and Discovery, Tunneling, IP within IP – Mobility Support in IPV6 – Mobility Header, Mobility Options, Dynamic Home Agent Address Discovery, Cache Management, Bidirectional Tunneling – TCP Over Wireless Networks – Indirect TCP -Snoop TCP – Mobile TCP.	Assessment: Internal Test Activity1: Online Certification Course – 13 to 15 Hours Linked in Learning. Preparatory Test	CO2	PO1, PO4,PO7,PO9
Module 3 (9Hours)	Module 3: OVERVIEW OF ANDROID Overview of Android What does Android run On – Android Internals? Android for mobile apps development Environment setup for Android apps Development Framework - Android- SDK, Eclipse Emulators – Emulator / Android AVD Android Emulation – Creation and set up First Android Application	Activity2: Case study Preparatory Test	CO3	PO1, PO2, PO3, PO4, PO5, PO7, PO8, PO9, , PO11
Module 4	Module 4: Android Activities		CO4	PO1, PO2, PO3, PO4,

(9Hours)	and GUI Design Concepts Design criteria for Android Application : Hardware Design Consideration, Design Demands For Android application, Intent, Activity, Activity Lifecycle and Manifest Creating Application and new Activities Simple UI - Layouts and Layout properties :Introduction to Android UI Design, Introducing Layouts XML Introduction to GUI objects viz.: Push Button , Text / Labels , EditText, ToggleButton , Padding ,	Activity2: case study Report writing Preparatory Test		PO5, PO7, PO8, PO9, , PO11,PO12
Module 5 (9Hours)	Module5: User Interface User Interface Design part 1: Views & View Groups, Views : Button, Text Field, Radio Button, Toggle Button, Checkbox, Spinner, Image View, Image switcher, Event Handling, Event Listeners, Layouts : Linear, Relative, List View, Grid View, Table View, Web View, Adapters. User Interface Design part 2: Menus, Action Bars, Notifications : Status, Toasts and Dialogs, Styles and Themes, Creating Custom Widgets, Focus, Touch Mode, Screen Orientation. Designing for Tablets – Working with tablets: Developing for different android platforms, Fragments, combining fragments into a multilane UI. Resources, Assets, Localization: Resources and Assets, Creating Resources, Managing application resources and assets, Resource-Switching in Android.	Assessment: Preparatory Exam Activity2: case study Report Submission and Viva	CO5	PO1, PO2, PO8, PO9, PO10, PO11,PO12

Text Books (TB):

TB1	Mobile Computing,Raj Kamal by Oxford,third edition, April 2018.
TB2	Mobile Computing Technology,Applications and service creation ,Asoke K Telukder, Roopa R Yavagal by TMH.
TB3	Dawn Griffiths and David Griffiths, “Head First Android Development”, 1st Edition, O’Reilly SPD Publishers, 2015.
TB4	J F DiMarzio, “Beginning Android Programming with Android Studio”, 4th Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126565580

Reference Books (RB):

RB1	Anubhav Pradhan, Anil V Deshpande, “ Composing Mobile Apps” using Android, Wiley 2014, ISBN: 978-81-265-4660-2
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Relevant Website (RW):

RW-1	Mobile Development Tutorials- Mobile Development Tutorials (tutorialspoint.com)
RW-2	Learn Android Studio Build Android Apps Quickly and Effectively- Learn Android Studio (barbra-coco.dyndns.org) .
RW-3	Andriod Studio - Learn Android Tutorial Android Studio Tutorial - Javatpoint
RW-4	Android For Developers: https://developer.android.com
RW-5	Add Firebase to your Android project https://firebase.google.com/docs/android/setup
RW-6	Android App Development in Android Studio- http://projanco.com
RW-7	How to develop an Android app- https:// www.educative.io/blog/how-to-develop-an-android-app
RW-8	Andriod Application Design with theme and styles - https://www.vogella.com/tutorials/AndroidStylesThemes/article.html

Video Links (VL):

VL-1	Introduction to Android Studio - https:// www.youtube.com/watch?v=bBt5sTXa0JA&list=PL49hKkxjHVqHilv8CUb-p9gMPFZtfqZTN&index=6
VL-2	Your First App https:// www.youtube.com/watch?v=2LnTEPPwucI&list=PL49hKkxjHVqHilv8CUb-p9gMPFZtfqZTN&index=7
VL-3	Android Development Environment - https:// www.youtube.com/watch?v=rjcpzn7ty0o&list=PL49hKkxjHVqHilv8CUb-p9gMPFZtfqZTN&index=10
VL-4	Application Fundamentals - https:// www.youtube.com/watch?v=Gf-MrhcOgnI&list=PL49hKkxjHVqHilv8CUb-p9gMPFZtfqZTN&index=12
VL-5	Activity LifeCycle I - https:// www.youtube.com/watch?v=DQxB18fzNnk&list=PL49hKkxjHVqHilv8CUb-p9gMPFZtfqZTN&index=14
VL-6	Intents - https:// www.youtube.com/watch?v=oLSyey1zszY&list=PL49hKkxjHVqHilv8CUb-p9gMPFZtfqZTN&index=21
VL-7	Saving Data - I - https://

	www.youtube.com/watch?v=oArUFc2mOHY&list=PL49hKkxjHVqHilv8CUB-p9gMPFZtfqZTN&index=28
VL-8	Security and Permissions - https:// www.youtube.com/watch?v=AvRhT6Jx-2Y&list=PL49hKkxjHVqHilv8CUB-p9gMPFZtfqZTN&index=30
VL-9	Services - https://www.youtube.com/watch?v=mhJKKAWZvgQ&list=PL49hKkxjHVqHilv8CUB-p9gMPFZtfqZTN&index=31

MOOC Courses (MC):LINKEDIN

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	LinkedIn Learning	5G Technology Strategy: Next-Generation Mobile Networking Welcome to 5G	CO1	https://www.linkedin.com/learning/5g-technology-strategy-next-generation-mobile-networking/welcome-to-5g?u=92695330	40 m
MC-2	LinkedIn Learning	Wireless Networking Essential Training Welcome to wireless networking	CO 2	https://www.linkedin.com/learning/wireless-networking-essential-training/welcome-to-wireless-networking?u=92695330	1hr 14m
MC-3	LinkedIn Learning	Android App Development : Working with Widgets	CO4	https://www.linkedin.com/learning/android-app-development-working-with-widgets/build-your-own-app-widget?u=92695330	1hr 39m
MC-4	LinkedIn Learning	Building an Android App with Architecture Components	CO4	https://www.linkedin.com/learning/building-an-android-app-with-architecture-components/welcome?u=92695330	2 hours 39 min
MC-5	LinkedIn Learning	Android Development Essential Training: 3 Navigation	CO4	https://www.linkedin.com/learning/android-development-essential-training-3-navigation/let-users-explore-your-app?u=92695330	1 hours 24 min
MC-6	LinkedIn Learning	Android App Development :	CO3	https://www.linkedin.com/learning/android-app-development-accessibility/welcome?u=92695330	2hr 56m

		Accessibility			
MC-7	LinkedIn Learning	Android Development Essential Training: 1 Your First App	CO3	https://www.linkedin.com/learning/android-development-essential-training-1-your-first-app/your-first-android-app?u=92695330	1hr 22m
MC-8	LinkedIn Learning	Android Studio Essential Training Android Studio and IntelliJ IDEA	CO5	https://www.linkedin.com/learning/android-studio-essential-training-14266841/android-studio-and-intellij-idea?u=92695330	2hr 50 m
MC-9	LinkedIn Learning	Android Development Essential Training: The user Interface with JAVA	CO5	https://www.linkedin.com/learning/android-development-essential-training-the-user-interface-with-kotlin/elements-of-an-android-user-interface?contextUrn=urn%3Ali%3AlyndaLearningPath%3A58ae2144498e686d31b4331a&u=92695330	1hr 53m
MC-10	LinkedIn Learning	<u>Android Development Essential Training: 2 User Interface Design</u>	CO5	https://www.linkedin.com/learning/android-development-essential-training-2-user-interface-design/user-interface-design?u=92695330	1hr 54m
Total MOOC Course integration with Certification					Approx 16 hours

MOOC:COURSE ERA

- 1.<https://www.coursera.org/learn/introduction-to-android-mobile-application-development>
- 2.<https://www.coursera.org/learn/get-started-with-android-app-development>

Hardware/Software/Equipment (HSE):

HSE-1	Java - https:// www.java.com/en/
HSE-2	Andriod studio by Google - https:// developer.android.com/studio

Session-wise Planning:

Module	Session	Topic	TB/RB/OR/RW/VL/	Pedagogy / Activity planned	CO	Mode of delivery
Bride Course	Zero Lecture	Mobile Computing- Need, Scope, Opportunity & Career	VL-1 to VL-5	Blended Learning Approach		Synchronous-PPT
I	1	Mobile handsets, Wireless Communications and Server Applications,	TB-1 ,TB2	PPT presentation & Classroom interaction	CO1	Synchronous-PPT Asynchronous-Video
	2	Cell phone System, Types of Telecommunication Networks	TB-1 ,TB2	PPT presentation & Classroom interaction		Interactive Lecture Synchronous-Video
	3	Components of wireless communication system	TB-1 ,TB2	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
	4	Architecture of mobile telecommunication system, wireless networking standards	TB-1 ,TB2	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
	6	Wireless LANs, Wireless LAN Architecture	TB-2 MC-1	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
	7	Applications of WLANs, Advantages of WLANs over wired LANs	TB-1 ,TB2	Flipped Class to Prepare comparative chart Advantages of WLANs over wired LANs		Synchronous-PPT Asynchronous-Video
	8	Bluetooth Technology, Protocol stack of Bluetooth.	TB-1 ,TB2	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video

	9	Activity-1	MC	Interactive & Assessment with Report		Assessment of Certification with Report
II	10	Mobile IP	TB1,TB2	PPT presentation & Classroom interaction	CO2	Synchronous-Interactive PPT
	11	Mobile Agent, Foreign Agent,	TB1,TB2	PPT presentation & Classroom interaction		Synchronous-Interactive PPT
	12	Care of Address, Registration,	TB1,TB2	PPT presentation & Classroom interaction		Synchronous - PPT
	13	Advertisement and Discovery, Tunneling,	TB1,TB2	PPT presentation & Classroom interaction		Synchronous - PPT
	14	IP within IP – Mobility Support in IPV6 – Mobility Header,	TB1,TB2,MC-2	PPT presentation & Classroom interaction		Synchronous – PPT Asynchronous - Video
	15	Mobility Options, Dynamic Home Agent Address Discovery,	TB1,TB2	PPT presentation & Classroom interaction		Synchronous-Interactive PPT
	16	Cache Management, Bidirectional	TB1,TB2	PPT presentation & Classroom interaction		Synchronous-Interactive PPT
	17	Tunneling – TCP Over Wireless Networks – Indirect TCP - Snoop TCP – Mobile TCP	TB1,TB2	PPT presentation & Classroom interaction		Synchronous-PPT

	18	Activity1	MC 2	Open & Close Discussion , Query Resolution , Monitorin g		Synchronous-Discussion, Suggestions Assessment Certification on Completion
III	19	Overview of Android	T3,T4,RB1 ,RW1,VL-1	PPT Presentati on & Demonstr ation	CO3	Synchronous-PPT Asynchronous-Video
	20-21	What does Android run On – Android Internals?	T3,T4,RB1	PPT Presentati on & Classroom interactio n		Synchronous-PPT Asynchronous-Video
	22-23	Android for mobile apps development Environment setup for Android apps	T3,T4,RB1,VL-3	PPT Presentati on & Demonstr ation		Synchronous-PPT Asynchronous-Video
	24	Development Framework - Android- SDK Eclipse Emulators – Emulator / Android AVD Android Emulation	T3,T4,RB1	PPT Presentati on		Synchronous-PPT Asynchronous-Video
	25	Creation and set up First Android Application	T3,T4,RB1 MC-7,RW2,VL-2	PPT Presentati on & Demonstr ation		Synchronous-PPT Asynchronous-Quiz
	26	ACTIVITY1	MC	Open & Close Discussion , Query Resolution , Monitorin g		Synchronous-Discussion, Suggestions Assessment Certification on Completion
	27	Activity-2- Case study Topic Allocation	MC	Interactiv e & Assessme nt with Report		Assessment Certification with Report

IV	28	Design criteria for Android Application : Hardware Design Consideration	T3,T4,RB1 MC-4,RW3,VL-4	PPT Presentation & Classroom interaction	CO4	Synchronous-PPT Lecture with interaction
	29	Design Demands For Android application, Intent, Activity,	T3,T4,RB1 VL6	PPT Presentation & Demonstration		Synchronous-PPT Lecture with demo
	30	Activity Lifecycle and Manifest Creating Application and new Activities	T3,T4,RB1,RW4,VL5	PPT Presentation & Classroom Interaction		Synchronous-PPT Asynchronous-Video
	31	Simple UI -Layouts and Layout properties	T3,T4,RB1 MC-3	PPT Presentation & Comparative Analysis		Synchronous-PPT Asynchronous-Video
	32	Introduction to Android UI Design, Introducing Layouts XML	T3,T4,RB1	Classroom interaction with live demo		Synchronous-PPT & Interactive Video with Quiz
	34	Introduction to GUI objects viz;Push Button , Text / Labels EditText, ToggleButton , Padding	T3,T4,RB1 MC5 VL7,RW5	PPT Presentation & Classroom Interaction		Synchronous-PPT Asynchronous-Video
	35	Activity1	MC	Interactive & Assessment with Report		Assessment Certification with Report
	36	Activity-2 Case study Report writing		Interactive & Assessment with Report		Synchronous-Video & Assessment Certification with Report

V	37	User Interface User Interface Design part 1: Views & View Groups, Views : Button, Text Field, Radio Button	T3,T4,RB1	PPT & Explanation	C05	Synchronous-PPT Presentation
	38	Toggle Button, Checkbox, Spinner, Image View, Image switcher	T3,T4,RB1	PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
	39	Layouts : Linear, Relative, List View, Grid View, Table View, Web View, Adapters.	T3,T4,RB1	Video Demonstration with Student Interaction		Synchronous-PPT
	40	User Interface Design part 2: Menus, Action Bars, Notifications : Status Toasts and Dialogs, Styles and Themes, Creating Custom Widgets, Focus, Touch Mode, Screen Orientation.	T3,T4,RB1 MC8,RW6 MC9	Video Demonstration with Student Interaction		Synchronous-PPT & Interactive Video with Quiz
	41	Designing for Tablets – Working with tablets: Developing for different android platforms	T3,T4,RB1 ,MC10 ,RW7,RW8 VL8	Video Demonstration with Student Interaction		Synchronous-PPT Asynchronous-Video
	42	Fragments, combining fragments into a multilane UI. Resources, Assets	T3,T4,RB1 MC-5 VL9	PPT & Video Demonstration with Student Interaction		Synchronous-PPT .
	43	Localization: Resources and Assets, Creating Resources, Managing application resources and assets, Resource-Switching in Android.	T3,T4,RB1	Synchronous interaction With students		Asynchronous-Video, PPT Assessment Certification on Completion
	44	ACTIVITY1	MC			

	45	Activity 2: Case Study Report Submission		Interactive & Assessment with Report		Asynchronous-Video, PPT Assessment Certification on Completion
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Assessment Scheme: IA: ESE - 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency	Weightage (%)	CO
1.	Activity 1	Formative	1	15	CO1,CO2, CO1,CO2,C O3 CO4,CO5
2.	Activity 2		1	15	CO1,CO2,CO3 CO4,CO5
3.	Internal Test		1	5	CO1, CO2
4.	Preparatory exam	Summative	1	10	CO1, CO2, CO3,CO4,CO 5
5.	Class participation		continuous	5	CO1, CO2, CO3,CO4,CO 5
6.	ESE	Summative	1	50	CO1, CO2, CO3,CO4,CO 5

Assessment Sheet with Rubrics for Grading & Evaluation

Activity-1 MOOC by LinkedIn Learning/Course era

[illegible]

Activity-2 Case Study on Mobile Application UI Design

Sr. No.	USN No.	Student Name	On-time Submission	Abstract Submission	Case Study Review	Report with Conclusion	Originality of Report (less than 12%)	Total	Conversion
			5 Marks	15 Marks	10 Marks	10 Marks	10 Marks	50 Marks	15Marks